Namma Space — Repository README & Reproduction Overview

Techfest IIT Bombay 2026–27 | Namma Space Challenge (Round 1)

Project Overview

Namma Space reconstructs photorealistic, navigable 3D digital twins of indoor real estate from standard smartphone walkthrough videos using 3D Gaussian Splatting (3DGS). The project delivers instant, zero-install first-person browser roaming with real-time telemetry and 60 FPS WebGL2 rendering.

Quick Start (Local & Cloud)

Option 1: Run Web Portal Locally:

```
git clone https://github.com/HemakshiAG/NammaSpace.git
cd NammaSpace
python serve_web.py
# Open: http://127.0.0.1:8080/index.html
```

Option 2: Cloud 3DGS Reconstruction (Google Colab):

Open notebooks/NammaSpace_Reconstruction_Colab.ipynb on Google Colab with T4 GPU. Step 1 compiles rasterizers with --no-build-isolation, Step 2 runs 30k iterations, and Step 3 downloads scene.splat and cameras.json.

First-Person Controls

- **W / S / A / D:** First-person forward, backward, left strafe, right strafe.
- **Mouse Left-Drag:** Orbit look orientation (pitch and yaw).
- **Mouse Right-Drag:** Lateral panning.
- **Mouse Scroll:** Field-of-view zoom.

Repository Structure

/web/: Static web portal, styles.css, viewer.js, scene.splat, cameras.json, _headers.
/docs/: Technical abstract, capture SOP, checklist, reconstruction report, QA matrix.
/notebooks/: 1-click Google Colab reconstruction notebook.
/test_data/: Sample walkthrough video dataset.

Upstream Attribution & Licences

Roomsplat (MIT), 3D Gaussian Splatting (Inria Non-commercial), COLMAP (BSD 3-Clause), GaussianSplats3D (MIT), Three.js (MIT).